

Program: Diploma in Artificial Intelligence	
Course Code: 3389	Course Title: MATLAB
Semester: 3	Credits: 0
Course Category: Program core course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To learn the features of MATLAB as a programming tool.
- Students will have an understanding of various programming constructs and how they can be used to solve a computational problem.
- Students will develop the skills to analyze and break down an engineering program and solve it algorithmically using MATLAB.

Course Prerequisites:

Topic	Course code	Course name	Semester
Sound knowledge of Python programming	10082	Introduction to IT systems Lab	I
Basic Computer programming knowledge	2139	Problem Solving and Programming Lab	II

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Installation of MATLAB and familiarization of various windows.	6	Applying
CO2	Learning basic programming concepts.	15	Applying
CO3	Implement programs using MATLAB functions and Graphics	15	Applying
CO4	Open ended experiment	6	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	2	2					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Installation of MATLAB and familiarization of various windows.		
M1.01	Installation of MATLAB	3	Applying
M1.02	Familiarization of MATLAB interface	3	Applying

CO2	Learning basic programming concepts.		
M2.01	Build and run MATLAB statements using commands.	3	Applying
M2.02	Develop simple MATLAB Programs using decision control structures	3	Applying
M2.03	Implement MATLAB Programs using loop control structures.	3	Applying
M2.04	Develop MATLAB Programs to perform basic operation in matrix	3	Applying
M2.05	Develop MATLAB Programs to solve algebraic equation	3	Applying
	Lab Exam – I	1.5	

CO3	Implement programs using MATLAB functions and Graphics		
M3.01	Develop MATLAB Programs using user defined functions and MATLAB functions	3	Applying
M3.02	Develop MATLAB Programs for drawing barchart and contours.	4	Applying
M3.03	Develop MATLAB program for plotting 2D and 3D	4	Applying
M3.04	Building MATLAB models using simulink	4	Applying
CO4	Open ended experiment		
M4.01	Open ended experiment**	6	Applying
	Lab Exam -2	1.5	

**** For example**

1.Practice some of these methods for accessing matrix elements with the following exercises:

- Create a matrix of size $N \times N$ that has ones in the border and zeros inside. For example if $N = 3$ the matrix can be created with

```
>> A=ones(3,3); A(2,2)=0
```

A =

```
1     1     1
1     0     1
1     1     1
```

Make this construction depend on N and work for any positive integer $N \geq 2$

- Create a 5x5 matrix whose rows are (1:5)
- Extract the diagonal of a given matrix without using diag (you may use size)
- Flip a given matrix horizontally. Vertically? Do not use fliplr or flipud
- Extract the anti-diagonal of a given matrix
- Extract the anti-diagonal, without first flipping it (Hint: use single index access)

2.Input two RGB images and perform arithmetic operations on it.

Text / Reference

T/R	Book Title/Author
T1	“A Guide to MATLAB, 3e: For Beginners and Experienced Users” Brian R. Hunt,Ronald L. Lipsman, Jonathan M. Rosenberg
R1	MATLAB for Engineers, 6th editionHolly Moore, Salt Lake Community CollegePearson Education Inc, 2022

Online Resources

Sl.No	Website Link
1	https://www.udemy.com
2	https://in.mathworks.com
3	https://www.coursera.org